Monotonic Sequences Worksheet — Solutions

Type A: The Difference Test

Use the difference $a_{n+1} - a_n$ to show that the given sequence $\{a_n\}$ is strictly increasing or strictly decreasing.

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1. $\left\{3 - \dfrac{2}{n}\right\}_{n=1}^{+\infty}$

$a_n = 3 - \dfrac{2}{n}$, so $a_{n+1} = 3 - \dfrac{2}{n+1}$.

$$a_{n+1} - a_n = \left(3 - \frac{2}{n+1}\right) - \left(3 - \frac{2}{n}\right) = \frac{2}{n} - \frac{2}{n+1} = \frac{2(n+1) - 2n}{n(n+1)} = \frac{2}{n(n+1)} > 0 \quad \text{for all } n \geq 1.$$

Since $a_{n+1} - a_n > 0$, we have $a_{n+1} > a_n$ for all n.

Answer: $\boxed{\text{strictly increasing}}$

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2. $\left\{\dfrac{n}{5n-2}\right\}_{n=1}^{+\infty}$

$a_n = \dfrac{n}{5n-2}$, so $a_{n+1} = \dfrac{n+1}{5(n+1)-2} = \dfrac{n+1}{5n+3}$.

$$\begin{aligned} a_{n+1} - a_n &= \frac{n+1}{5n+3} - \frac{n}{5n-2} = \frac{(n+1)(5n-2) - n(5n+3)}{(5n+3)(5n-2)} \\ &= \frac{(5n^2 + 3n - 2) - (5n^2 + 3n)}{(5n+3)(5n-2)} = \frac{-2}{(5n+3)(5n-2)}. \end{aligned}$$

For $n \geq 1$, both $5n + 3 > 0$ and $5n - 2 > 0$, so $a_{n+1} - a_n < 0$, i.e. $a_{n+1} < a_n$.

Answer: $\boxed{\text{strictly decreasing}}$

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3. $\{n - 3^n\}_{n=1}^{+\infty}$

$a_n = n - 3^n$, so $a_{n+1} = (n+1) - 3^{n+1}$.

$$a_{n+1} - a_n = \left[(n+1) - 3^{n+1}\right] - \left[n - 3^n\right] = 1 - 3^{n+1} + 3^n = 1 - 3 \cdot 3^n + 3^n = 1 - 2 \cdot 3^n.$$

For $n \geq 1$, $2 \cdot 3^n \geq 6 > 1$, so $a_{n+1} - a_n < 0$, i.e. $a_{n+1} < a_n$.

Answer: $\boxed{\text{strictly decreasing}}$

Type B: The Ratio Test

Use the ratio a_{n+1}/a_n to show that the given sequence $\{a_n\}$ is strictly increasing or strictly decreasing. All terms of these sequences are positive.

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4. $\left\{\dfrac{3^n}{2+3^n}\right\}_{n=1}^{+\infty}$

$a_n = \dfrac{3^n}{2+3^n}$, so $a_{n+1} = \dfrac{3^{n+1}}{2+3^{n+1}}$.

$$\frac{a_{n+1}}{a_n} = \frac{3^{n+1}}{2+3^{n+1}} \cdot \frac{2+3^n}{3^n} = \frac{3(2+3^n)}{2+3^{n+1}} = \frac{6+3^{n+1}}{2+3^{n+1}}.$$

The numerator is larger than the denominator ($6 > 2$), so $\dfrac{a_{n+1}}{a_n} > 1$, i.e. $a_{n+1} > a_n$.

Answer: strictly increasing

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5. $\left\{\dfrac{n}{3^n}\right\}_{n=1}^{+\infty}$

$a_n = \dfrac{n}{3^n}$, so $a_{n+1} = \dfrac{n+1}{3^{n+1}}$.

$$\frac{a_{n+1}}{a_n} = \frac{n+1}{3^{n+1}} \cdot \frac{3^n}{n} = \frac{n+1}{3n}.$$

For $n \geq 1$: $n+1 \leq 2n < 3n$, so $\dfrac{a_{n+1}}{a_n} = \dfrac{n+1}{3n} < 1$, i.e. $a_{n+1} < a_n$.

Answer: strictly decreasing

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6. $\left\{\dfrac{(2n)!}{(n!)^2}\right\}_{n=1}^{+\infty}$

$a_n = \dfrac{(2n)!}{(n!)^2}$, so $a_{n+1} = \dfrac{(2n+2)!}{((n+1)!)^2}$. Using $(2n+2)! = (2n+2)(2n+1)\,(2n)!$ and $(n+1)! = (n+1)\,n!$:

$$\frac{a_{n+1}}{a_n} = \frac{(2n+2)(2n+1)\,(2n)!}{(n+1)^2\,(n!)^2} \cdot \frac{(n!)^2}{(2n)!} = \frac{(2n+2)(2n+1)}{(n+1)^2} = \frac{2(n+1)(2n+1)}{(n+1)^2} = \frac{2(2n+1)}{n+1} = \frac{4n+2}{n+1}.$$

Since $4n+2 > n+1$ for $n \geq 1$, $\dfrac{a_{n+1}}{a_n} > 1$, i.e. $a_{n+1} > a_n$.

Answer: strictly increasing